Theoretical Knowledge
Advanced Log Analysis
1.Log Correlation
Log correlation involves combining logs from multiple sources (e.g., system logs, network logs) to identify attack patterns.

Example:
Failed login attempts (Event ID 4625) followed by outbound traffic may indicate compromise.

2. Anomaly Detection
Anomaly detection identifies unusual patterns such as:

- Logins at odd hours
- High data transfer
- Sudden spikes in activity

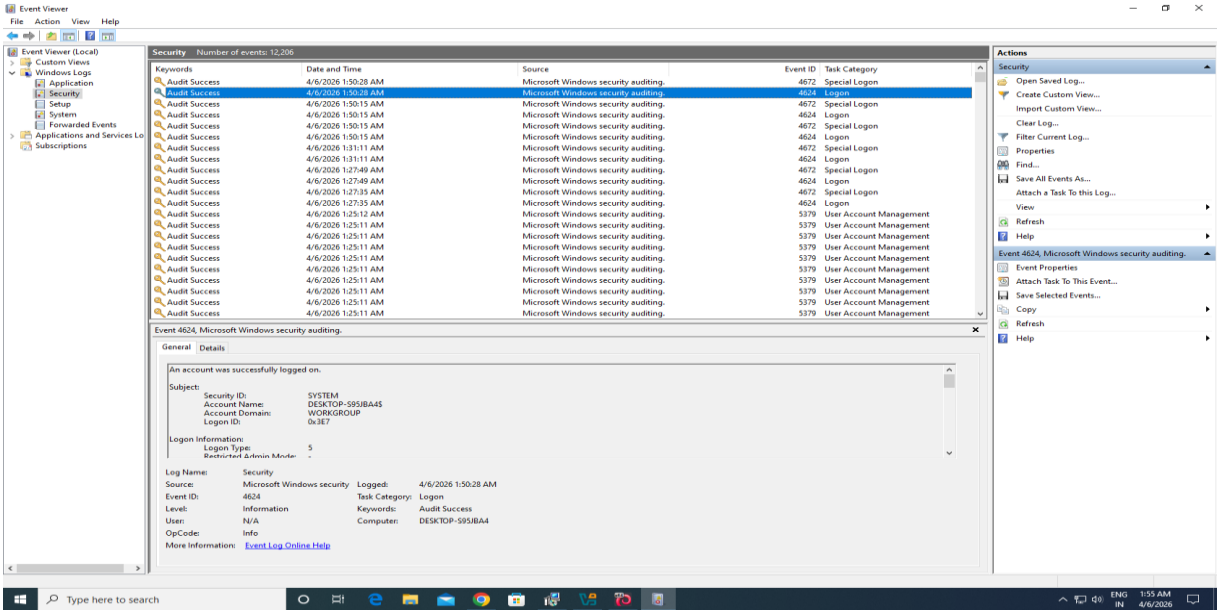
3.Log Enrichment
Log enrichment adds context like:

- IP geolocation
- User roles
- Threat intelligence data

Threat Intelligence Integration
1.Indicators of Compromise (IOCs)
Malicious IP addresses
File hashes
Domains

2.TTPs (Tactics, Techniques, Procedures)
Using frameworks like MITRE ATT&CK to understand attacker behavior.

3.Threat Hunting
Proactively searching logs for suspicious patterns such as:
Unauthorized access (T1078)



Incident Escalation

SOC Tiers

Tier 1: Monitoring & triage

Tier 2: Investigation

Tier 3: Advanced response

Conclusion:

A structured communication used to report incidents clearly to stakeholders.